

2024 年秋季学期期末试卷答案

一、1. ✓ 2. ✓ 3. ×

4. B 5. D 6. B

7. $f(w_2) = 6w_2(1 - w_2)$ (注: w_2 可换为任意变量), 8:00:30

8. $\lambda\mu, \lambda(\mu^2 + \sigma^2)$

9. $\frac{37}{75}, \frac{1}{2}$

二、(1) 以 9 点为时间起点, 以小时为时间单位, 用过程 $N(t)$ 表示到 t 时刻为止访问商店的顾客人数, 可知 $N(t)$ 为非齐次泊松过程, 强度 $\lambda(t) = \begin{cases} 2, & 0 \leq t < 1 \\ 3, & 1 \leq t \leq 2 \end{cases}$. 则 9:30 到 10:30 间没有顾客访问的概率为

$$\begin{aligned} P(N(1.5) - N(0.5) = 0) &= P(N(1.5) - N(1) = 0)P(N(1) - N(0.5) = 0) \\ &= e^{-\int_1^{1.5} 3du} e^{-\int_{0.5}^1 2du} = e^{-2.5} \end{aligned}$$

(2) 到 11 点为止, 有 2 位以上顾客访问的概率为

$$P(N(2) \geq 2) = 1 - P(N(2) = 0) - P(N(2) = 1)$$

其中

$$\begin{aligned} P(N(2) = 0) &= P(N(2) - N(1) = 0)P(N(1) = 0) \\ &= e^{-\int_1^2 3du} e^{-\int_0^1 2du} = e^{-5} \end{aligned}$$

$$\begin{aligned} P(N(2) = 1) &= P(N(2) - N(1) = 1)P(N(1) = 0) + P(N(2) - N(1) = 0)P(N(1) = 1) \\ &= \int_1^2 3du e^{-\int_1^2 3du} e^{-\int_0^1 2du} + e^{-\int_1^2 3du} \int_0^1 2du e^{-\int_0^1 2du} = 5e^{-5} \end{aligned}$$

故而 $P(N(3) \geq 2) = 1 - 6e^{-5}$

(3) 运用书本定理 2.1 可知, 对齐次泊松过程 $N'(t)$, 给定 $N'(t) = 1$, 则过程第一个事件的到达时刻 $W_1' \sim U[0, t]$.

注意到

$$\begin{aligned} &\{N(1.25) - N(0.75) = 1\} \\ &= \{N(1.25) - N(1) = 1, N(1) - N(0.75) = 0\} \cup \{N(1.25) - N(1) = 0, N(1) - N(0.75) = 1\}. \end{aligned}$$

记 $\{N(1.25) - N(1) = 1, N(1) - N(0.75) = 0\} = A$, $\{N(1.25) - N(1) = 0, N(1) - N(0.75) = 1\} = B$, A 与 B 为互斥事件. $P(A) = 3 * 0.25e^{-3*0.25} = 0.75e^{-0.75}$, $P(B) = 2 * 0.25e^{-2*0.25} = 0.5e^{-0.5}$.

记 W_1 为过程在 9:45 后第一个事件的到达时刻, 则本小题所求分布密度函数为

$$\begin{aligned} & f(W_1 = w_1|A \cup B) \\ &= f(W_1 = w_1|A)P(A|A \cup B) + f(W_1 = w_1|B)P(B|A \cup B) \\ &= f(W_1 = w_1|A) \frac{0.75e^{-0.75}}{0.75e^{-0.75} + 0.5e^{-0.5}} + f(W_1 = w_1|B) \frac{0.5e^{-0.5}}{0.75e^{-0.75} + 0.5e^{-0.5}} \\ &= \begin{cases} \frac{3e^{-0.75}}{0.75e^{-0.75} + 0.5e^{-0.5}}, & 1 \leq w_1 \leq 1.25 \\ \frac{2e^{-0.5}}{0.75e^{-0.75} + 0.5e^{-0.5}}, & 0.75 \leq w_1 \leq 1 \end{cases} \end{aligned}$$

可知该分布为一个分段均匀分布.

三、(1) 由状态转移图 (免) 易证该马氏链是不可约, 且每个状态都是非周期正常返的, 因而是遍历的.

(2) 求解线性方程组:

$$\begin{cases} \pi_0 = 0.5\pi_0 + 0.25\pi_2 + 0.25\pi_3 \\ \pi_1 = 0.5\pi_0 + 0.25\pi_1 + 0.25\pi_3 \\ \pi_2 = 0.75\pi_1 + 0.5\pi_2 + 0.25\pi_3 \\ \pi_3 = 0.25\pi_2 + 0.25\pi_3 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 = 1 \end{cases}$$

得平稳分布 (极限分布): $\pi = (\frac{6}{23}, \frac{5}{23}, \frac{9}{23}, \frac{3}{23})$.

(3) 当初始分布 $\pi_0 = (\frac{6}{23}, \frac{5}{23}, \frac{9}{23}, \frac{3}{23})$ 时, 该马氏链为严平稳序列. 此时:

$$EX_n = EX_0 = \frac{32}{23}, \quad \text{Var}(X_n) = \text{Var}(X_0) = \frac{540}{529}$$

四、(1) $R_Y(\tau) = EY(t+\tau)Y(t) = EX(t+\tau)X(t)\cos(2\pi(t+\tau) + \Theta)\cos(2\pi t + \Theta)$

$$= R_X(\tau)E\cos(2\pi(t+\tau) + \Theta)\cos(2\pi t + \Theta)$$

$$= R_X(\tau) \frac{1}{2\pi} \int_0^{2\pi} \cos(2\pi(t+\tau) + \theta)\cos(2\pi t + \theta)d\theta$$

$$= R_X(\tau) \frac{1}{4\pi} \int_0^{2\pi} [\cos(2\pi(2t+\tau) + 2\theta) + \cos(2\pi\tau)]d\theta$$

$$= \frac{1}{2} R_X(\tau) \cos(2\pi\tau) = e^{-|\tau|} \cos(2\pi\tau)$$

(2) 现求 $Y(t)$ 的功率谱密度函数, 由维纳——辛钦公式

$$\begin{aligned}
S_Y(\omega) &= \int_{-\infty}^{+\infty} R_Y(\tau) e^{-j\omega\tau} d\tau = 2 \int_0^{+\infty} e^{-\tau} \cos(2\pi\tau) \cos(\omega\tau) d\tau \\
&= \int_0^{+\infty} e^{-\tau} [\cos((2\pi + \omega)\tau) + \cos((2\pi - \omega)\tau)] d\tau \\
&= \int_0^{+\infty} e^{-\tau} \cos((2\pi + \omega)\tau) d\tau + \int_0^{+\infty} e^{-\tau} \cos((2\pi - \omega)\tau) d\tau
\end{aligned}$$

现计算 $\int_0^{+\infty} e^{-\tau} \cos((2\pi + \omega)\tau) d\tau$

$$\begin{aligned}
\int_0^{+\infty} e^{-\tau} \cos((2\pi + \omega)\tau) d\tau &= \frac{1}{2\pi + \omega} \int_0^{+\infty} e^{-\tau} d\sin((2\pi + \omega)\tau) \\
&= \frac{1}{2\pi + \omega} \int_0^{+\infty} e^{-\tau} \sin((2\pi + \omega)\tau) d\tau \\
&= -\frac{1}{(2\pi + \omega)^2} \int_0^{+\infty} e^{-\tau} d\cos((2\pi + \omega)\tau) \\
&= \frac{1}{(2\pi + \omega)^2} - \frac{1}{(2\pi + \omega)^2} \int_0^{+\infty} e^{-\tau} \cos((2\pi + \omega)\tau) d\tau
\end{aligned}$$

因此,

$$\int_0^{+\infty} e^{-\tau} \cos((2\pi + \omega)\tau) d\tau = \frac{1}{1 + (2\pi + \omega)^2}.$$

同样的方法可求得

$$\int_0^{+\infty} e^{-\tau} \cos((2\pi - \omega)\tau) d\tau = \frac{1}{1 + (2\pi - \omega)^2}.$$

因此,

$$S_Y(\omega) = \frac{1}{1 + (2\pi + \omega)^2} + \frac{1}{1 + (2\pi - \omega)^2}$$

五、(1) $EX_0 = \int_0^1 2x^2 dx = \frac{2}{3}$, 对于 $n \geq 1$ 有

$$EX_n = E[E(X_n|X_{n-1})] = E\left[\int_{1-x_{n-1}}^1 \frac{x_n}{x_{n-1}} dx_n\right] = E\left[1 - \frac{1}{2}X_{n-1}\right] = 1 - \frac{1}{2}EX_{n-1}$$

因为 $EX_0 = \frac{2}{3}$, 递推后有 $EX_n = \frac{2}{3}$

(2) $R_X(n, n+m) = E\left(X_n - \frac{2}{3}\right)\left(X_{n+m} - \frac{2}{3}\right) = E(X_n X_{n+m}) - \frac{4}{9}$, 现求 $E(X_n X_{n+m}) - \frac{4}{9}$

$$\begin{aligned}
E(X_n X_{n+m}) &= E[E(X_n X_{n+m}|X_n)] = E[X_n E(X_{n+m}|X_n)] = E\left[X_n \left(1 - \frac{1}{2}E(X_{n+m-1}|X_n)\right)\right] \\
&= EX_n - \frac{1}{2}E[E(X_n X_{n+m-1}|X_n)] = \frac{2}{3} - \frac{1}{2}E(X_n X_{n+m-1})
\end{aligned}$$

因此,

$$E(X_n X_{n+m}) - \frac{4}{9} = -\frac{1}{2}\left(E(X_n X_{n+m-1}) - \frac{4}{9}\right) = \cdots = \left(-\frac{1}{2}\right)^m \left(EX_n^2 - \frac{4}{9}\right)$$

现求 EX_n^2 , $EX_0^2 = \int_0^1 2x^3 dx = \frac{1}{2}$, 对于 $n \geq 1$ 有

$$EX_n^2 = E[E(X_n^2|X_{n-1})] = E\left[\int_{1-x_{n-1}}^1 \frac{x_n^2}{x_{n-1}} dx_n\right] = 1 - EX_{n-1} + \frac{1}{3}EX_{n-1}^2$$

因为 $EX_0 = \frac{2}{3}$, $EX_0^2 = \frac{1}{2}$, 递推后有 $EX_n^2 = \frac{1}{2}$

因此, $R_X(n, n+m) = E(X_n X_{n+m}) - \frac{4}{9} = \left(-\frac{1}{2}\right)^m \left(EX_n^2 - \frac{4}{9}\right) = \frac{1}{18} \left(-\frac{1}{2}\right)^m = R(m)$

(3) 由宽平稳过程的定义知 $\{X_n\}$ 为宽平稳过程, 因为 $\lim_{m \rightarrow \infty} R(m) = 0$, 该平稳过程有均值遍历性